A forward quadratic on a flat-boundary kernel: structural conclusions for the exact B-adaPG controller

Abstract

We analyze the original undamped, full-ambient B-adaPG method on an oscillatory Legendre kernel that is constant on a boundary edge, with a quadratic tangential term placed entirely in the forward objective. For every interior initialization and every pair of positive initial stepsizes, cumulative steps diverge, the normal coordinate has an exact cumulative-step asymptotic, and the squared tangential coordinate is summable with outgoing-step weights. The unique minimizer is a cluster point, and every subsequence with outgoing steps bounded away from zero converges to it. Boundary-uniform relative smoothness genuinely fails, although compact-interior relative smoothness holds. These are structural partial conclusions: neither full point or objective convergence, nor a divergent orbit, nor eventual selection of a controller branch is established. A finite high-precision illustration is separated from the proofs. No novelty claim is made.

Keywords: adaptive Bregman proximal gradient; flat boundary kernel; forward quadratic; cumulative step sizes; weighted summability; boundary asymptotics

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1 Kernel, objective, and exact initialization

For a differentiable convex function v, write

$$D_v(u,z) = v(u) - v(z) - \langle \nabla v(z), u - z\rangle, \qquad \Delta_v(u,z) = D_v(u,z) + D_v(z,u).$$

The second argument is always in the differentiability domain. Let $h^*(p) = \sup_z\{\langle p, z\rangle - h(z)\}$ denote the full ambient conjugate. Set

$$e = 1/256, \qquad a = 1/100, \qquad K = [-1,1]\times[0,e], \qquad C = \operatorname{int} K.$$

Using continuous endpoint values and $0\log 0 = 0$, define

$$\begin{aligned} b(x) &= (1+x)\log(1+x) + (1-x)\log(1-x),\\ q(y) &= -\sqrt{y} - \sqrt{e-y},\\ h(x,y) &= y\{b(x) - ax\sin(\log y)\} + q(y) \quad ((x,y)\in K,\ y>0),\\ h(x,0) &= -\sqrt{e} \quad (-1\le x\le 1), \qquad h = +\infty \text{ off } K. \end{aligned} \tag{1}$$

For any $\kappa > 0$, place all tangential objective curvature in the forward slot:

$$f(x,y) = y + \kappa x^2, \qquad g = 0.$$

The constrained minimum over K is 0, attained uniquely at $z_* = (0,0)$. The Hessian of f is $\mathrm{diag}(2\kappa, 0)$; in particular, this objective is not strongly convex on the ambient plane.

We use the original B-adaPG update and controller (2.3), (2.6) of Ou, Latafat and Themelis [1], pinned to version 2. Choose arbitrary $z_0 = (x_0, y_0) \in C$ and arbitrary finite $\gamma_0, \gamma_1 > 0$. The first primal update uses γ_1. For $k \geq 0$,

$$z_{k+1} = \operatorname*{argmin}_{z \in K} \left\{ \langle \nabla f(z_k), z - z_k \rangle + \frac{D_h(z, z_k)}{\gamma_{k+1}} \right\}, \qquad P_k = \nabla h(z_k). \tag{2}$$

For $k \geq 1$ put

$$\rho_k = \gamma_k/\gamma_{k-1}, \qquad r_{k+1} = \sqrt{1+\rho_k}, \qquad \delta_{k+1} = 2r_{k+1}, \qquad Q_k = \Delta_h(z_k, z_{k-1}),$$

$$\alpha_k = \frac{D_h(z_k, z_{k-1})}{D_h(z_{k-1}, z_k)}, \qquad \ell_k = \frac{2\kappa(x_k - x_{k-1})^2}{Q_k}, \qquad H_k = \nabla h - \gamma_k \nabla f.$$

The actual ambient probe and selector specialize to

$$\begin{aligned} v_k &= \nabla h^*(P_k - \delta_{k+1}\gamma_k(2\kappa x_k, 1)), \\ \Lambda_k &= \frac{2D_h(z_k, v_k)}{\delta_{k+1}^2 Q_k}, \\ \rho_{k+1} &= \min\left\{ r_{k+1}, \frac{\alpha_k}{(1+\alpha_k)2r_{k+1}[\Lambda_k - (1-\gamma_k\ell_k)]_+} \right\}, \\ \gamma_{k+1} &= \gamma_k \rho_{k+1}. \end{aligned} \tag{3}$$

Here $[t]_+ = \max\{t, 0\}$ and positive divided by zero is $+\infty$. There is no projection, damping, imposed stepsize floor, curve restriction, or prescribed stepsize schedule.

For completeness, the inverse map can be described without a two-dimensional minimization. Given $p = (p_x, p_y) \in \mathbb{R}^2$, solve

$$x(y) = \tanh\left(\frac{p_x/y + a\sin(\log y)}{2} \right), \qquad b(x(y)) - ax(y)[\sin(\log y) + \cos(\log y)] + q'(y) = p_y. \tag{4}$$

The solution lies in $0 < y < e$, and $\nabla h^*(p) = (x(y), y)$. Its uniqueness and global existence follow below; the derivative of the scalar left-hand side is the positive Hessian Schur complement.

2 Structural partial theorem

Theorem 1. *For every initialization in Section 1, the complete run is well defined, $z_k \in C$, and every generated stepsize is finite and positive. Let $S_k = \sum_{j=1}^{k} \gamma_j$. Then*

$$P_k^x = P_0^x - 2\kappa \sum_{j=1}^{k} \gamma_j x_{j-1}, \qquad P_k^y = P_0^y - S_k, \tag{5}$$

$$S_k \longrightarrow +\infty, \qquad y_k \longrightarrow 0, \qquad S_k^2 y_k \longrightarrow \tfrac{1}{4}, \tag{6}$$

$$\sum_{k=2}^{\infty} \gamma_{k+1} x_k^2 < \infty. \tag{7}$$

Moreover $\inf_{k\geq 0} f(z_k) = 0$*, and* z_* *is a cluster point. For each* $c > 0$*, along every sequence of indices* $k \to \infty$ *satisfying* $\gamma_{k+1} \geq c$*, one has* $z_k \to z_*$*.*

The source's compact-interior relative-smoothness and Bregman-zone hypotheses hold. However, no finite relative-smoothness modulus works on $C \cap B$ *for any open ball* B *containing a point* $(\xi, 0)$ *with* $|\xi| < 1$*.*

The theorem does not assert $x_k \to 0$ for the full sequence, convergence of the full objective sequence, or eventual selection of either controller branch. It is neither a full convergence theorem nor a counterexample for this family. The general unique-minimizer question under compact-interior relative smoothness remains open here.

3 Kernel and smoothness assumptions

Write $s = \sin(\log y)$, $c = \cos(\log y)$, and $B = b''(x) = 2/(1-x^2)$. The interior derivatives are

$$\begin{aligned} h_x &= y[b'(x) - as], & h_y &= b(x) - ax(s+c) + q'(y), \\ h_{xx} &= yB, & h_{xy} &= b'(x) - a(s+c), \\ h_{yy} &= q''(y) - ax(c-s)/y, & q''(y) &= \tfrac{1}{4}[y^{-3/2} + (e-y)^{-3/2}]. \end{aligned}$$

With $u = \operatorname{atanh} x$, $b' = 2u$ and $B = 2\cosh^2 u$. Since $\cosh u \geq 1 + u^2/2 \geq \sqrt{2}|u|$, one has $b'^2/B \leq 1$. As $B \geq 2$, this yields

$$\frac{|h_{xy}|}{\sqrt{B}} \leq 1 + a, \qquad y h_{yy} \geq 4 - \sqrt{2}a.$$

The Schur complement therefore satisfies

$$T := h_{yy} - \frac{h_{xy}^2}{h_{xx}} \geq \frac{4 - \sqrt{2}a - (1+a)^2}{y} > 0. \tag{8}$$

Thus $\nabla^2 h$ is positive definite on C. Continuity of b on $[-1,1]$ and $y\sin(\log y) \to 0$ establish continuity of h on K. Interior convexity extends to boundary chords by continuity. Its infinity extension is proper, closed, and convex. Boundedness of its domain gives 1-coercivity, in the sense $h(z)/\|z\| \to +\infty$ as $\|z\| \to \infty$.

The component h_y diverges uniformly in x at either horizontal edge, because b and $ax(s+c)$ are bounded. At either vertical edge away from the horizontal edges, h_x diverges. The gradient norm thus diverges at every boundary approach. No boundary point has a finite subgradient: at the top and bottom, inward vertical difference quotients tend to $-\infty$; at the remaining side points the same holds for inward horizontal quotients. This

covers corners using the vertical direction. Consequently $\operatorname{dom}\partial h = C$. Strict interior convexity now gives essential strict convexity, and steepness gives essential smoothness. Hence h is Legendre despite its flat bottom edge. Standard conjugacy for a 1-coercive Legendre function implies that h^* is finite and continuously differentiable on $\mathbb{R}^2$, with $\nabla h : C \to \mathbb{R}^2$ a bijection and inverse ∇h^*. The positive definite Hessian and inverse-function theorem give a continuous positive definite Hessian for h^* as well.

We verify the two boundary-zone requirements directly. For fixed $w = (X, Y) \in K$ and $z = (x, y) \to w$ from C, continuity reduces $D_h(w, z) \to 0$ to

$$h_x(x, y)(X - x) \to 0, \qquad h_y(x, y)(Y - y) \to 0.$$

For the first, use bounded y and $(X - x)b'(x) \to 0$; at a vertical endpoint the latter follows from $(1 - |x|)|b'(x)| \to 0$. For the second, the nonsingular terms are bounded and $(Y - y)q'(y) \to 0$ at either square-root endpoint, as well as at interior Y. These estimates include mixed corners. Thus Assumption 2.4(i) of [1] holds. Every reverse-divergence sublevel in its second argument is bounded because C is bounded, proving Assumption 2.4(ii). No converse implication from small boundary divergence to point convergence is used.

The function f is a finite convex polynomial on all of $\mathbb{R}^2$, and $g = 0$ is finite closed convex. On each compact convex subset of C, the smallest eigenvalue of $\nabla^2 h$ has a positive lower bound. A finite L then makes $L\nabla^2 h - \operatorname{diag}(2\kappa, 0)$ positive semidefinite, which is precisely compact-interior relative smoothness.

In contrast, let B_0 be an open ball containing $(\xi, 0)$, $|\xi| < 1$. For all sufficiently small $y > 0$, $(\xi, y) \in C \cap B_0$. Convexity of $Lh - f$ on this open convex set would require

$$Lh_{xx}(\xi, y) - f_{xx}(\xi, y) = \frac{2Ly}{1 - \xi^2} - 2\kappa \geq 0.$$

No finite L satisfies this as $y \downarrow 0$. Thus the boundary-uniform hypothesis of the companion boundary-uniform convergence result does not cover this example. The quadratic has not been moved into the proximal term.

4 Well-posedness and exact dual recurrence

For any finite positive γ, the primal problem is minimization of h plus an affine function. The properties just proved give a unique minimizer in C, with

$$\nabla h(z_{k+1}) = \nabla h(z_k) - \gamma_{k+1}(2\kappa x_k, 1). \tag{9}$$

Telescoping proves (5). The second dual component changes strictly at every step, so consecutive primal points differ. Both directed divergences in α_k are strictly positive. The conjugate is defined at every ambient probe, and its inverse gradient is interior. The

selector takes the minimum of a finite positive growth candidate and a positive or infinite second candidate. Induction proves well-posedness for all time.

To verify the specialized probe rather than assuming it, (9) gives

$$H_k(z_k) - H_k(z_{k-1}) = -\gamma_k \nabla f(z_k).$$

The general ambient argument $P_k + \delta_{k+1}[H_k(z_k) - H_k(z_{k-1})]$ is therefore exactly the argument in (3). Conjugate Bregman duality gives

$$D_{h^*}(P_k - \delta_{k+1}\gamma_k \nabla f(z_k), P_k) = D_h(z_k, v_k).$$

Finally $\Delta_f(z_k, z_{k-1}) = 2\kappa(x_k - x_{k-1})^2$. These identities verify every specialized controller formula, including the order of the divergence and the gradient evaluation at z_k.

5 Retained energy and tangential summability

We give the algebraic energy calculation explicitly. It needs no boundary-uniform relative smoothness. For $w \in K$, set $G_j(w) = f(z_j) - f(w)$. For $k \geq 2$ define

$$d_k = 1 - \frac{\rho_k}{2r_k}, \qquad U_k(w) = D_h(w, z_k) + \gamma_k(1 + r_k)G_{k-1}(w) + d_k D_h(z_k, z_{k-1}).$$

No controller bound is imposed on the arbitrary initial ratio ρ_1. The first branch gives $\rho_k \leq r_k$ and $d_k \geq 1/2$ for $k \geq 2$. Put $\vartheta = r_{k+1}$,

$$T_k = (1 - \gamma_k \ell_k)Q_k, \qquad E_{k+1} = \rho_{k+1}\langle H_k(z_k) - H_k(z_{k-1}), z_{k+1} - z_k\rangle.$$

The primal relation and three-point identity imply

$$D_h(w, z_{k+1}) + \gamma_{k+1}G_k(w) + D_h(z_{k+1}, z_k) = D_h(w, z_k) + E_{k+1} - \gamma_{k+1}D_f(w, z_k),$$
$$G_{k-1}(w) - G_k(w) = T_k/\gamma_k + D_f(z_{k-1}, z_k).$$

Adding $\gamma_{k+1}\vartheta$ times the second identity to the first yields

$$\begin{aligned} &D_h(w, z_{k+1}) + \gamma_{k+1}(1 + \vartheta)G_k(w) + D_h(z_{k+1}, z_k) \\ &= D_h(w, z_k) + \gamma_{k+1}\vartheta G_{k-1}(w) - \rho_{k+1}\vartheta T_k + E_{k+1} \\ &\quad - \gamma_{k+1}\{D_f(w, z_k) + \vartheta D_f(z_{k-1}, z_k)\}. \end{aligned}$$

Fenchel's inequality in Bregman form is

$$\langle v, u - z\rangle \leq D_h(u, z) + D_{h^*}(\nabla h(z) + v, \nabla h(z)).$$

Apply it with $v = \delta_{k+1}[H_k(z_k) - H_k(z_{k-1})]$ and multiply by ρ_{k+1}/δ_{k+1}. The exact probe then gives

$$E_{k+1} \leq \frac{\rho_{k+1}}{\delta_{k+1}} D_h(z_{k+1}, z_k) + \frac{\delta_{k+1}\rho_{k+1}}{2}\Lambda_k Q_k.$$

Using $\delta_{k+1} = 2r_{k+1}$ and $Q_k = (1+\alpha_k)D_h(z_k, z_{k-1})/\alpha_k$, collect coefficients to obtain

$$\begin{aligned} U_{k+1}(w) \le U_k(w) - A_k G_{k-1}(w) - B_k D_h(z_k, z_{k-1}) \\ - \gamma_{k+1} D_f(w, z_k), \end{aligned} \tag{10}$$

where the omitted term is $-\gamma_{k+1} r_{k+1} D_f(z_{k-1}, z_k) \le 0$, and

$$A_k = \gamma_k(1 + r_k) - \gamma_{k+1} r_{k+1},$$

$$B_k = d_k - \rho_{k+1} r_{k+1} \frac{1+\alpha_k}{\alpha_k}[\Lambda_k - (1 - \gamma_k \ell_k)].$$

The bracket in B_k is signed. Since $\rho_{k+1} r_{k+1} \le r_{k+1}^2 = 1 + \rho_k$, one has $A_k \ge \gamma_k(r_k - \rho_k) \ge 0$. If the bracket is nonpositive, $B_k \ge d_k$; if positive, the second controller branch bounds the subtracted term by $1/2$. Thus $B_k \ge 0$ in both cases.

At $w = z_*$, all objective gaps are nonnegative, $U_k(z_*) \ge 0$, and $D_f(z_*, z_k) = \kappa x_k^2$. Telescoping (10) gives the quantitative bound

$$\kappa \sum_{k=2}^{N} \gamma_{k+1} x_k^2 \le U_2(z_*) \qquad (N \ge 2). \tag{11}$$

This proves (7). The energy argument is the retained smooth-defect form of the source's main identity and Bregman Young inequality; it uses no boundary point-separation assertion.

6 Cumulative steps and normal asymptotics

Suppose for contradiction that $S_\infty < \infty$. Because $|x_k| \le 1$, both series in (5) converge absolutely. Thus P_k converges in $\mathbb{R}^2$. Continuity of ∇h^* confines z_k to a compact subset of C, and $\gamma_k \to 0$. The growth bound $\rho_{k+1} \le \sqrt{1 + \rho_k}$ implies

$$\rho_k \le R := \max\{\rho_1, (1+\sqrt{5})/2\}, \qquad 1 \le r_{k+1} \le R.$$

Consequently the dual probes also lie in a compact neighborhood for all sufficiently large k.

Here is a uniform Taylor verification of the controller limit. Write $F_k = \nabla f(z_k)$ and $M_k = \nabla^2 h^*(P_k)$. These matrices are uniformly positive definite and continuous on a compact dual neighborhood; F_k is bounded and $\|F_k\| \ge 1$. By (9), $P_{k-1} = P_k + \gamma_k F_{k-1}$ and $z_k - z_{k-1} = O(\gamma_k)$. Since f has Lipschitz gradient, $F_{k-1} - F_k = O(\gamma_k)$. Uniform integral Hessian formulas and Taylor expansion give

$$\frac{Q_k}{\gamma_k^2} = F_k^T M_k F_k + o(1), \qquad \alpha_k \to 1,$$

$$\frac{2D_{h^*}(P_k - \delta_{k+1}\gamma_k F_k, P_k)}{\delta_{k+1}^2 \gamma_k^2} = F_k^T M_k F_k + o(1).$$

Uniformity in the second expression follows from $2 \le \delta_{k+1} \le 2R$. The leading quadratic form is bounded below by a positive constant. Hence $\Lambda_k \to 1$. On the compact primal neighborhood $Q_k \ge \mu\|z_k - z_{k-1}\|^2$ for some $\mu > 0$, so $0 \le \ell_k \le 2\kappa/\mu$ and $\gamma_k \ell_k \to 0$.

The second ratio candidate in (3) now tends to $+\infty$, while the growth candidate stays bounded. Eventually, under this contradictory finite-total-step hypothesis only,

$$\gamma_{k+1} = \gamma_k \sqrt{1 + \rho_k} > \gamma_k.$$

This contradicts $\gamma_k \to 0$, proving $S_k \to +\infty$.

The explicit vertical derivative is

$$P_k^y = b(x_k) - a x_k[\sin(\log y_k) + \cos(\log y_k)] - \frac{1}{2\sqrt{y_k}} + \frac{1}{2\sqrt{e - y_k}}. \tag{12}$$

Its first two terms are uniformly bounded. Since $P_k^y = P_0^y - S_k \to -\infty$, necessarily $y_k \to 0$: away from zero, the right-hand side is bounded below, including near e where its singular term is positive. Once $y_k \to 0$, the last positive term is bounded, and (12) gives

$$\frac{1}{2\sqrt{y_k}} = S_k + O(1).$$

Therefore $S_k\sqrt{y_k} \to 1/2$ and $S_k^2 y_k \to 1/4$. No monotonicity of y_k is asserted.

7 Optimal clusters and the unresolved tangential question

The tail $\sum_{k\ge2} \gamma_{k+1}$ diverges. Combining this with (7) proves $\liminf_{k\to\infty} x_k^2 = 0$: an eventual positive lower bound would make the weighted series diverge. Select indices $k_j \to \infty$ with $x_{k_j} \to 0$. Since already $y_k \to 0$, it follows that

$$z_{k_j} \to (0, 0), \qquad f(z_{k_j}) \to 0.$$

Nonnegativity of f on K gives $\inf_k f(z_k) = 0$; equivalently, the running best values $\min_{0\le j\le N} f(z_j)$ tend to zero. Furthermore, convergence of the nonnegative series gives $\gamma_{k+1} x_k^2 \to 0$. On any subsequence with $\gamma_{k+1} \ge c > 0$, this forces $x_k \to 0$, hence $z_k \to z_*$. This completes the proof of Theorem 1.

The distinction between incoming and outgoing steps is essential: (7) controls the old endpoint z_k with its outgoing weight γ_{k+1}. It need not control x_k at indices where this weight tends to zero. The first branch bounds a ratio only from above. Meanwhile, near $x = 0$, tangential kernel curvature is $h_{xx} \sim 2y_k \sim 1/(2S_k^2)$, whereas forward curvature remains 2κ. Small absolute steps can therefore still be large on that tangential scale. A full convergence theorem would need an actual-controller estimate excluding repeated tangential excursions at increasingly small heights. A counterexample would need an

exact initialized orbit realizing such excursions. Neither is supplied by these identities or by the finite experiment below. In particular, the temporary eventual-growth conclusion used inside the contradiction in Section 6 is not a statement about the actual infinite-time run.

A Noncertified finite numerical illustration

The accompanying reproducibility files evaluate 100 updates for $\kappa = 1$, $z_0 = (0.002, 0.0001)$, and $\gamma_0 = \gamma_1 = 0.0001$. They record states 0 through 100 and controller evaluations 1 through 100; controller 100 computes a prospective next stepsize, but update 101 is not taken. Update 1 uses the initialized stepsize.

The accompanying `mpmath` implementation solves (4) by bracketed scalar Newton iterations using the Schur derivative. It evaluates the general ambient H_k formula and checks conjugate divergence against reversed-primal divergence. Constants are parsed as arbitrary-precision decimals or rationals, but computations are approximate and are not interval-certified.

Independent runs at 100 and 130 decimal digits agree on every recorded branch. Both have 86 growth selections and 14 curvature selections. Curvature is selected at

$$8,\ 17,\ 18,\ 26,\ 27,\ 36,\ 37,\ 47,\ 56,\ 66,\ 74,\ 83,\ 92,\ 93.$$

For orientation, the 130-digit run gives

$$x_{100} \approx -4.28200052647111 71\,10^{-7}, \qquad y_{100} \approx 9.98826963592701 82\,10^{-5},$$

$$\gamma_{100} \approx 2.01539138414350 98\,10^{-4}.$$

All recorded states are interior and all recorded steps are positive. Agreement across precisions is a reproducibility check, not a certified error bound and not evidence establishing any infinite-time claim.

An earlier double-precision exploration is withdrawn: its first different controller branch is at index 65, and its final state materially differs. Failed outputs are excluded. Cancellation in tiny divergences is a conditioning concern, not a proved explanation of every discrepancy. No numerical quantity is used in a proof; this experiment establishes no asymptotic convergence, divergence, summability, or eventual branch choice.

References

[1] H. Ou, P. Latafat and A. Themelis. Linesearch-free adaptive Bregman proximal gradient for convex minimization under local relative smoothness. *arXiv:2508.01353v2*, 5 June 2026. `https://arxiv.org/abs/2508.01353v2`.